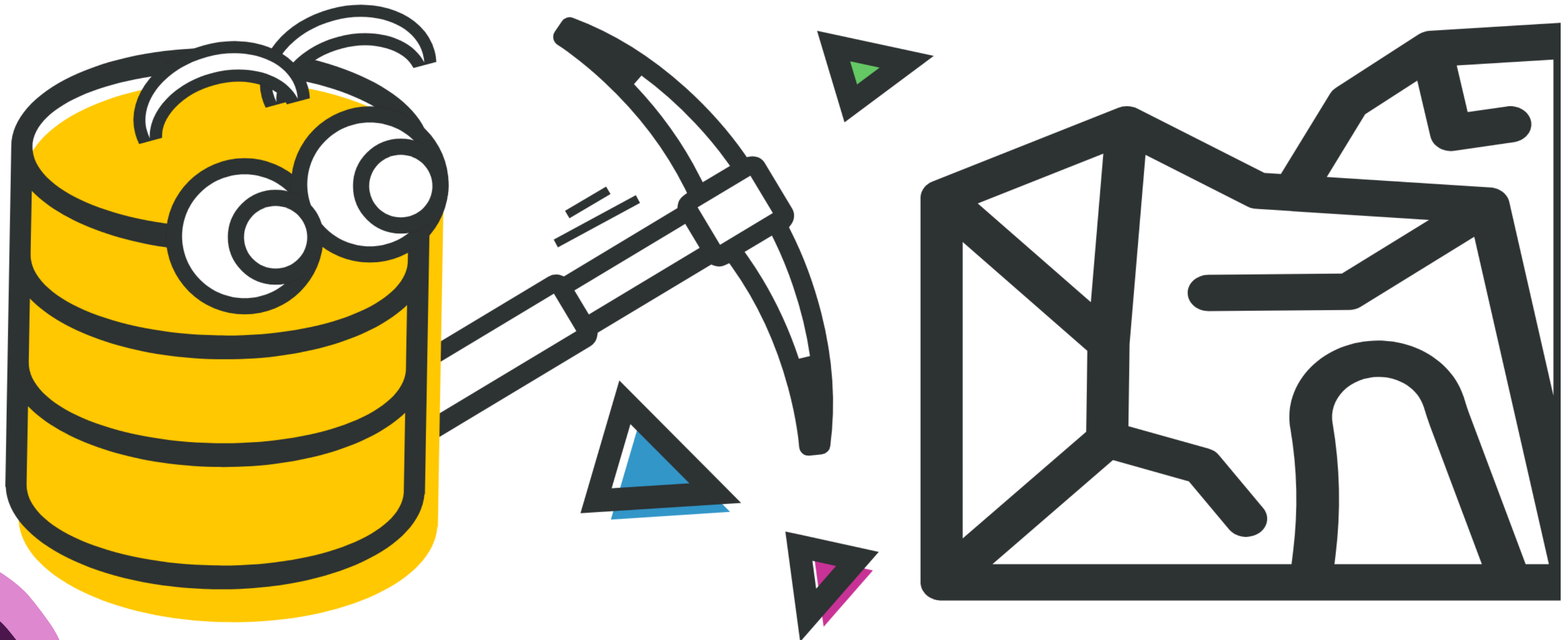


Data Mining

Understanding Data for Better Decisions



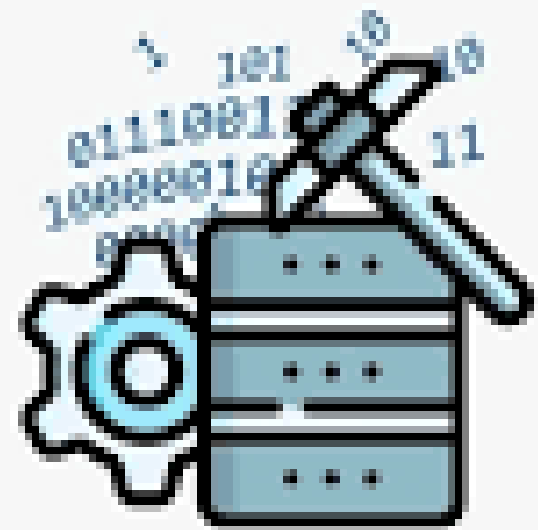
Group 7: Lucky Seven

Nguyen Minh Nguyen

Tran Dang Khoa

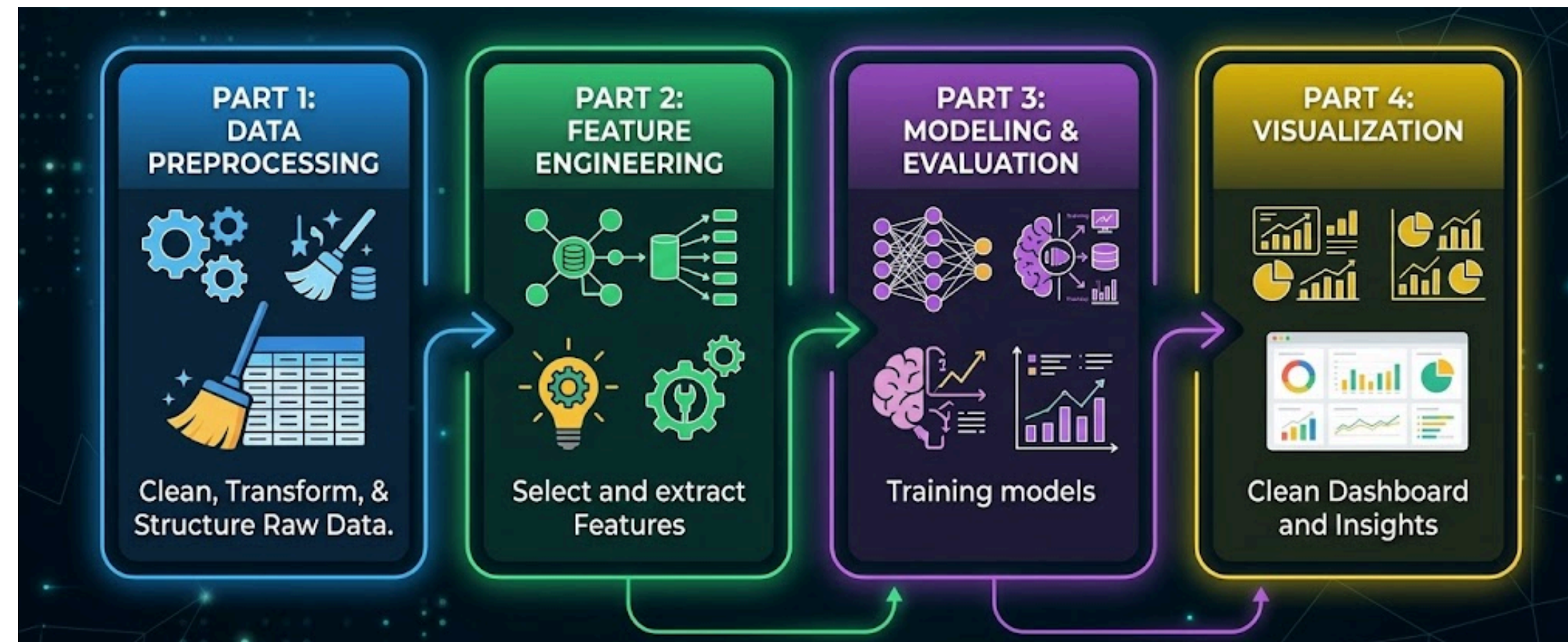
Nguyen Gia Kiet

Nguyen Hoang Anh Khoa



Data Mining

WorkFlow



Part 1: Data Preprocessing

Building a Clean Dataset from the Olist E-Commerce Ecosystem

Objective: Ingest, clean, and consolidate fragmented data into a unified, model-ready dataset (`cleaned_orders.csv`)

□ Data Quality Check — `cleaned_orders.csv`

Scope: Processing 6 core tables from the Olist e-commerce ecosystem, covering 99K+ raw orders

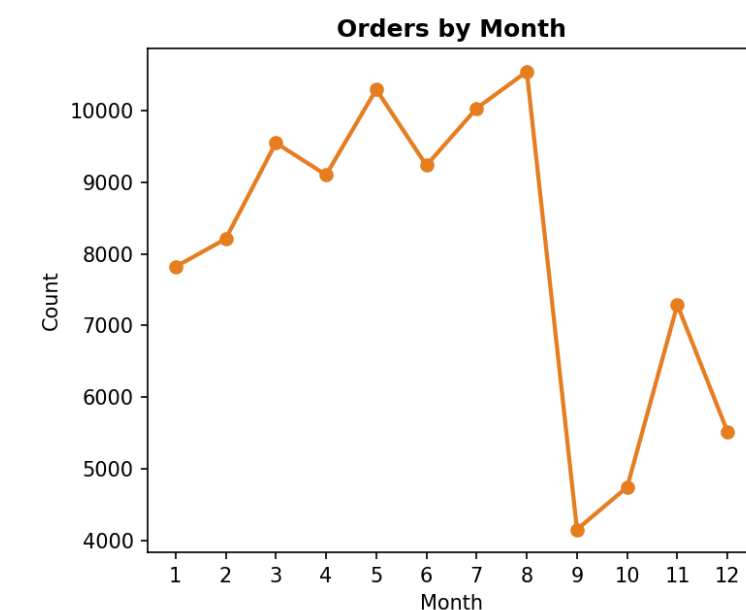
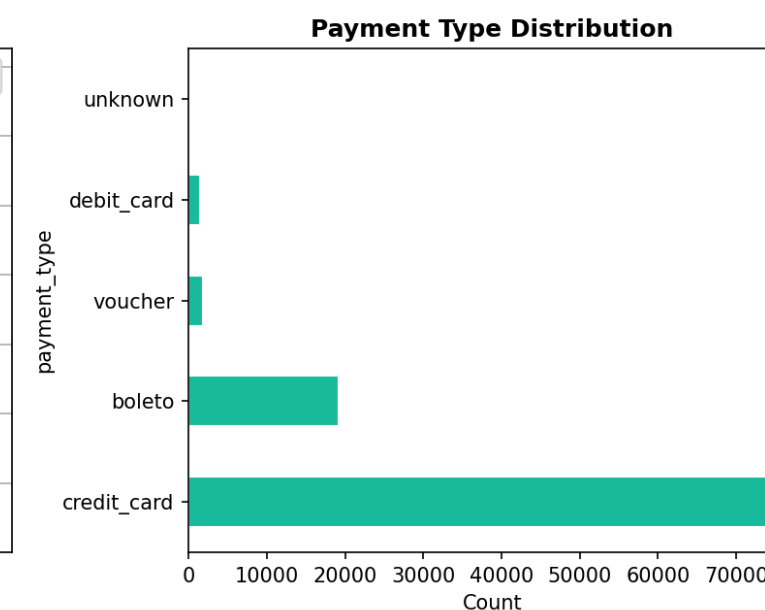
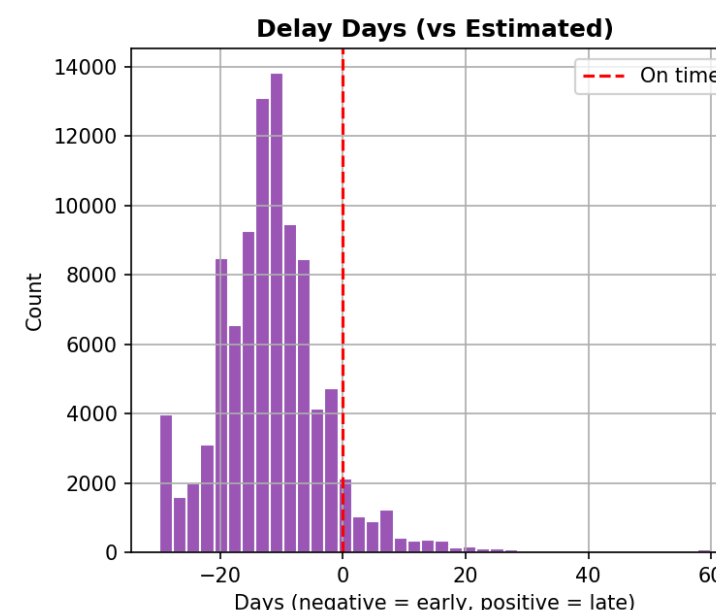
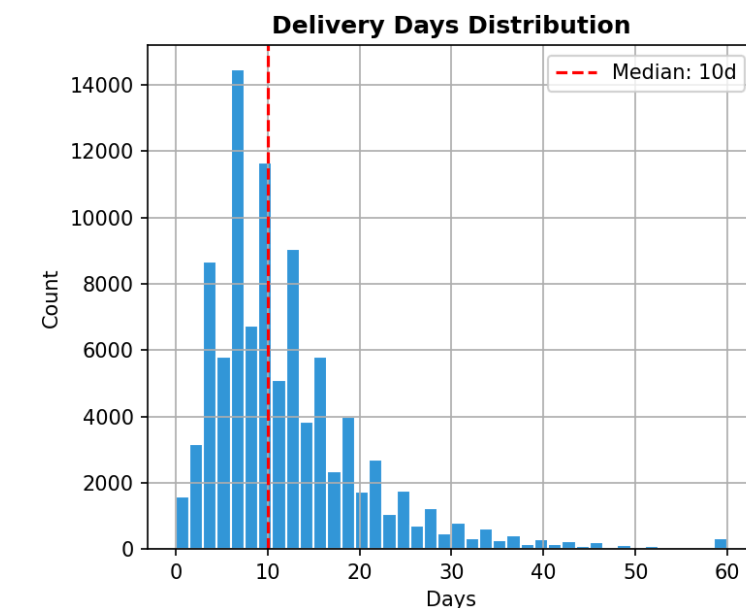
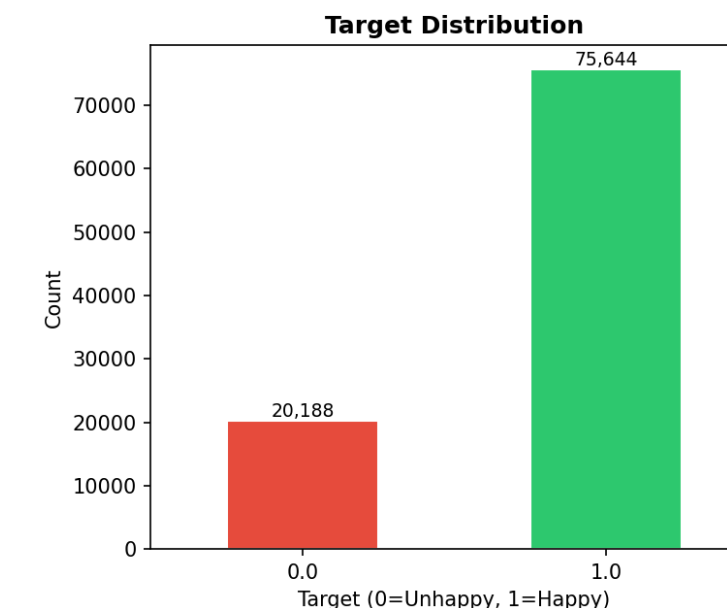
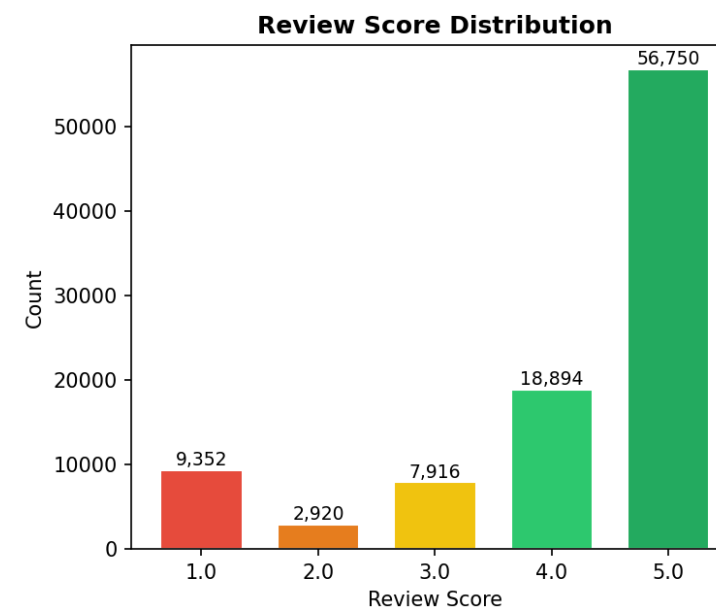
Key Pipeline:

Filtering valid orders

→ Merging relational tables

→ Imputing missing values

→ Labeling the target variable



Part 1.1: Load Datasets & Initial Filter

Raw Data Ingestion: Successfully loaded 6 core relational tables from the Olist ecosystem

Dataset Shapes: Combined total of over 99K+ original orders across different files

The Filtering Rule: Applied strict rule `order_status == 'delivered'` to target only completed transactions

Final Filter Result:

Retained: 96,478 valid orders (97.0% of the dataset)

Removed: 2,963 anomalous/uncompleted orders (canceled, unavailable, processing...)

Dataset shapes:

```
-----  
orders      : 99,441 rows x 8 cols  
reviews     : 99,224 rows x 7 cols  
items       : 112,650 rows x 7 cols  
payments    : 103,886 rows x 5 cols  
products    : 32,951 rows x 9 cols  
customers   : 99,441 rows x 5 cols
```

Phân phối order_status trước khi lọc:

```
order_status  
delivered    96478  
shipped      1107  
canceled     625  
unavailable  609  
invoiced     314  
processing   301  
created       5  
approved      2  
Name: count, dtype: int64
```

Tổng số đơn hàng: 99,441

Part 1.2: Merge & Solve Missing Value

Table Merging: Executed LEFT JOIN operations using orders_delivered as the base table to prevent data loss

Handling Duplicates & Items: Retained the latest review record for each order, and aggregated multiple items per order to a single row using sum, max, and mode functions

Missing Value Imputation:

Filled empty review comments with a blank string "

Imputed missing product physical dimensions with column Median values

Excluded entries with missing customer geographical locations via dropna

Result: A fully consolidated master dataframe containing 96,478 rows and 29 baseline features

```
Sơ đồ merge:
orders_delivered
├── LEFT JOIN reviews ON order_id
├── LEFT JOIN items ON order_id → LEFT JOIN products ON product_id
├── LEFT JOIN payments ON order_id (aggregate: sum & mode)
└── LEFT JOIN customers ON customer_id
```

Missing Values Report:			
	Missing Count	Missing %	Dtype
review_comment_title	85286	88.40	str
review_comment_message	57576	59.68	str
avg_product_description_len	1332	1.38	float64
product_category	1332	1.38	str
avg_product_photos_qty	1332	1.38	float64
review_score	646	0.67	float64
review_creation_date	646	0.67	str
avg_product_height_cm	16	0.02	float64
avg_product_weight_g	16	0.02	float64
avg_product_width_cm	16	0.02	float64
avg_product_length_cm	16	0.02	float64
order_approved_at	14	0.01	str
order_delivered_customer_date	8	0.01	str
order_delivered_carrier_date	2	0.00	str
payment_type	1	0.00	str
payment_installments	1	0.00	float64
total_payment_value	1	0.00	float64

Part 1.3: Target Labeling & Final Dataset

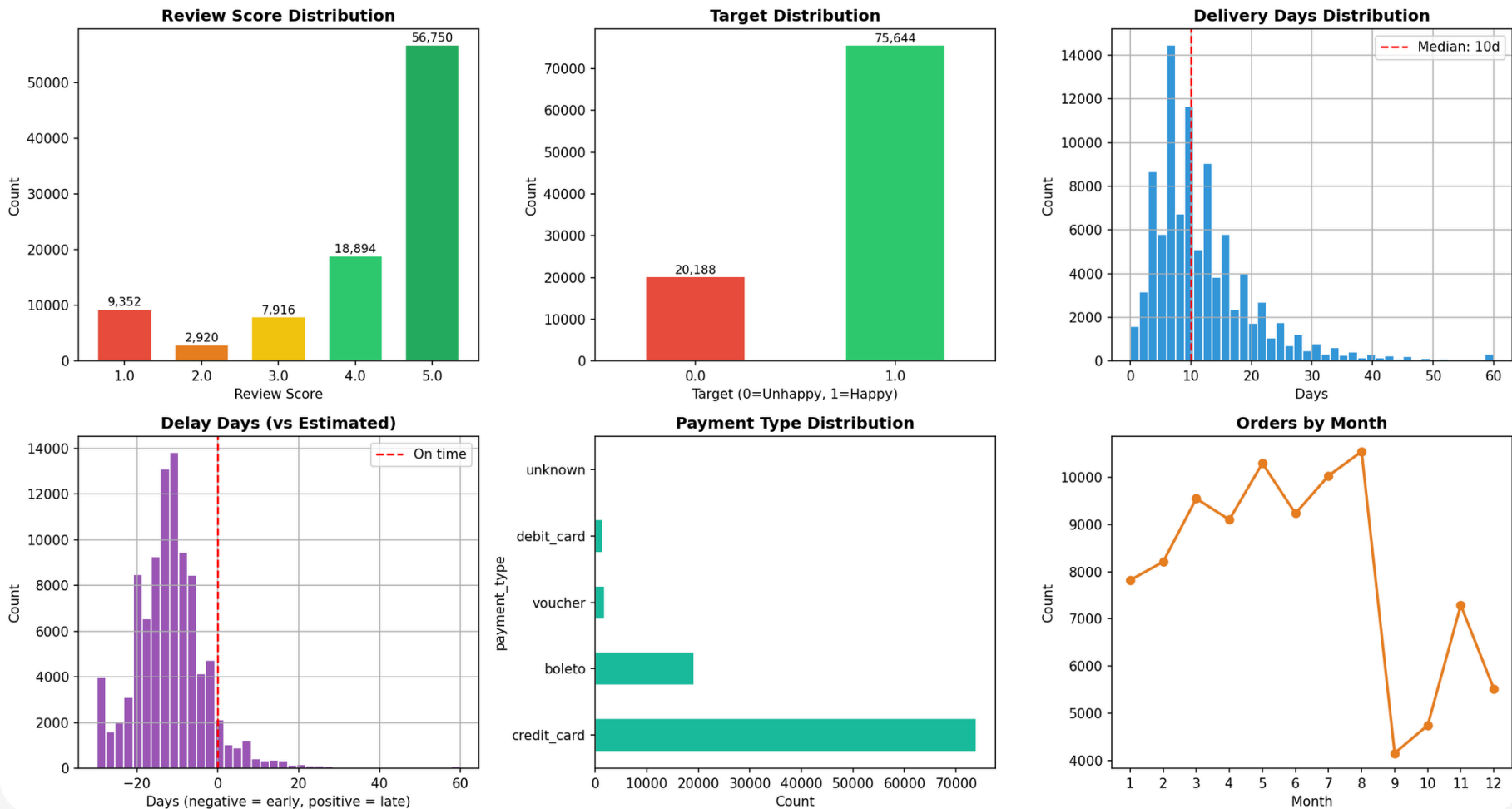
Target Formulation: Grouped review_score into binary classes:
Scores 1-3 → Target = 0 (Dissatisfied); Scores 4-5 → Target = 1 (Satisfied)

Class Distribution: High class imbalance detected with 75.6K Satisfied orders (78.4%) vs 20.1K Dissatisfied orders (20.9%)

Quick Data Insights: Credit card is the most dominant payment method. The average actual delivery time is 12.1 days

Dataset Export: Exported cleaned_orders.csv containing 96,478 total delivered records with an average payment value of 159.85 BRL

Data Quality Check — cleaned_orders.csv



```
Phân phối target variable:
Hài lòng (1): 75,644 (78.4%)
Không hài lòng (0): 20,188 (20.9%)
Chưa có review (NaN): 646 (0.7%)

Class ratio (0:1) = 0.21 : 0.79
Class imbalance khá cao – cân nhắc dùng SMOTE hoặc class_weight khi train
```

PREPROCESSING SUMMARY REPORT

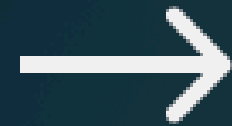
```
-----
Total delivered orders      : 96,478
Orders có review (trainable) : 95,832
Orders không có review    : 646
Target = 1 (hài lòng)      : 75,644
Target = 0 (không hài lòng) : 20,188
Avg delivery days          : 12.1 ngày
Avg delay days             : -11.9 ngày
Avg total payment value    : 159.85
```

Part 2: Feature Engineering

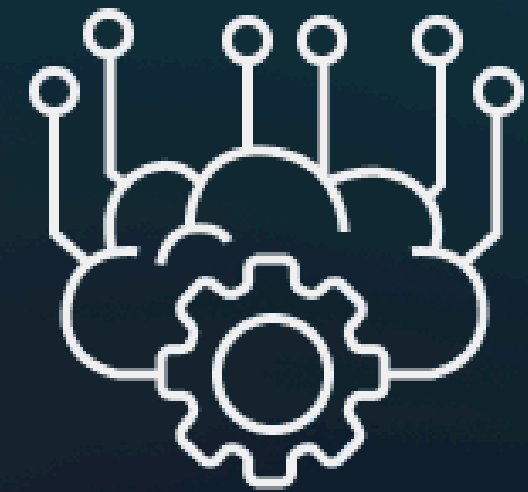
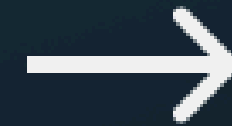
Larana, Inc.



Raw data



Feature
Engineering



ML model

Objective:

Transform raw data into meaningful features to improve model performance

Input:

cleaned_orders.csv

Output:

featured_orders.csv

Load cleaned data

```
df = pd.read_csv(
    "../data/processed/cleaned_orders.csv"
)

df.head()
```

✓ 3.2s Python

	order_id	customer_id	customer_unique_id	order_status	order_purchase_timestamp	order_approved_at	order_delivered_carrier_date
0	e481f51cbdc54678b7cc49136f2d6af7	9ef432eb6251297304e76186b10a928d	7c396fd4830fd04220f754e42b4e5bff	delivered	2017-10-02 10:56:33	2017-10-02 11:07:15	2017-10-04 19:55:00
1	53cdb2fc8bc7dce0b6741e2150273451	b0830fb4747a6c6d20dea0b8c802d7ef	af07308b275d755c9edb36a90c618231	delivered	2018-07-24 20:41:37	2018-07-26 03:24:27	2018-07-26 14:31:00
2	47770eb9100c2d0c44946d9cf07ec65d	41ce2a54c0b03bf3443c3d931a367089	3a653a41f6f9fc3d2a113cf8398680e8	delivered	2018-08-08 08:38:49	2018-08-08 08:55:23	2018-08-08 13:50:00
3	949d5b44dbf5de918fe9c16f97b45f8a	f88197465ea7920adcdbec7375364d82	7c142cf63193a1473d2e66489a9ae977	delivered	2017-11-18 19:28:06	2017-11-18 19:45:59	2017-11-22 13:39:59
4	ad21c59c0840e6cb83a9ceb5573f8159	8ab97904e6daea8866dbdbc4fb7aad2c	72632f0f9dd73dfee390c9b22eb56dd6	delivered	2018-02-13 21:18:39	2018-02-13 22:20:29	2018-02-14 19:46:34

5 rows × 38 columns

Shipping Ratio

```
df['shipping_ratio'] = (
    df['total_freight_value']
    / df['total_order_value']
)

df[['shipping_ratio']].head()
```

✓ 0.0s

	shipping_ratio
0	0.225265
1	0.160894
2	0.107302
3	0.376731
4	0.304682

Late Delivery Flag

```
df['is_late_delivery'] = np.where(
    df['delay_days'] > 0,
    1,
    0
)

df[['is_late_delivery']].head()
```

✓ 0.0s

	is_late_delivery
0	0
1	0
2	0
3	0
4	0

Save Featured Dataset

Generate Code Markdown

```
df.to_csv(
    "../data/processed/featured_orders.csv",
    index=False
)

print("Feature engineering completed successfully!")
```

✓ 4.4s

Feature engineering completed successfully!

Delivery & Shipping Features

Feature	Purpose
total_order_value	Total order value
shipping_ratio	Shipping fee ratio
is_late_delivery	Late delivery indicator
expensive_shipping	High shipping cost

Delivery experience affects customer satisfaction

Product Features

Feature	Purpose
product_volume_cm3	Product volume
is_heavy_product	Heavy product indicator
product_info_score	Product information quality

Product Volume = Length × Width × Height

Customer Behavior Features

Feature	Purpose
is_installment	Installment payment
is_weekend_purchase	Weekend purchase

Customer behavior influences expectations

Feature Engineering Result

- Created meaningful features
- Handled missing values
- Prepared dataset for modeling

Output:

featured_orders.csv

Goal:

Improve prediction performance

Save Featured Dataset

```
df.to_csv(
    "../data/processed/featured_orders.csv",
    index=False
)

print("Feature engineering completed successfully!")
```

✓ 4.4s

Feature engineering completed successfully!

df.head()
✓ 0.2s
Python

	order_id	customer_id	customer_unique_id	order_status	order_purchase_timestamp	order_approved_at	order_delivered_carrier_date
0	e481f51cbdc54678b7cc49136f2d6af7	9ef432eb6251297304e76186b10a928d	7c396fd4830fd04220f754e42b4e5bff	delivered	2017-10-02 10:56:33	2017-10-02 11:07:15	2017-10-04 19:55:00
1	53cdb2fc8bc7dce0b6741e2150273451	b0830fb4747a6c6d20dea0b8c802d7ef	af07308b275d755c9edb36a90c618231	delivered	2018-07-24 20:41:37	2018-07-26 03:24:27	2018-07-26 14:31:00
2	47770eb9100c2d0c44946d9cf07ec65d	41ce2a54c0b03bf3443c3d931a367089	3a653a41f6f9fc3d2a113cf8398680e8	delivered	2018-08-08 08:38:49	2018-08-08 08:55:23	2018-08-08 13:50:00
3	949d5b44dbf5de918fe9c16f97b45f8a	f88197465ea7920adcdbec7375364d82	7c142cf63193a1473d2e66489a9ae977	delivered	2017-11-18 19:28:06	2017-11-18 19:45:59	2017-11-22 13:39:59
4	ad21c59c0840e6cb83a9ceb5573f8159	8ab97904e6daea8866dbdbc4fb7aad2c	72632f0f9dd73dfee390c9b22eb56dd6	delivered	2018-02-13 21:18:39	2018-02-13 22:20:29	2018-02-14 19:46:34

5 rows × 44 columns

Part 3 : Modeling & Evaluation

Removing errors, duplicates, inconsistencies, and missing values to ensure data becomes accurate, reliable, and ready for analysis.

Data Cleaning

- **Objective: Early Warning System for At-Risk Customers (Class 0)**

- **Core Evaluation Metric: Optimizing Recall (Negative)**

- **Core Philosophy: "Better to over-alert for proactive customer care than to miss a dissatisfied customer."**

Transforming raw data into structured formats through organizing, encoding, grouping, and preparing it for deeper

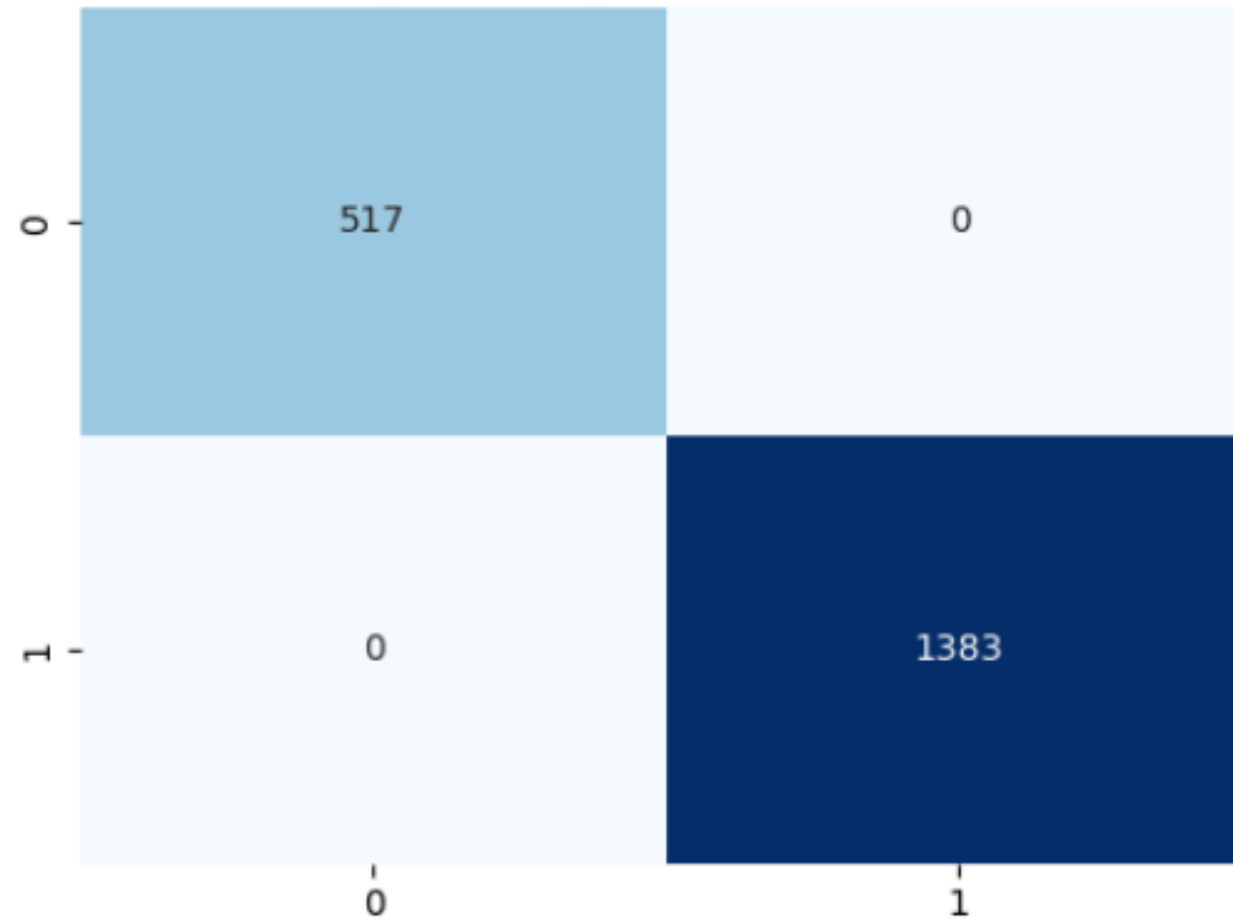
**Fixing Data
Leakage**

- **The Issue: Initial models showed 100% accuracy (Illusion)**
- **Root Cause: Models learned from leaked post-purchase features (review_score, has_review)**
- **The Solution: Completely removed review-related data. Forced models to learn exclusively from operational metrics (delivery days, freight value, etc.)**

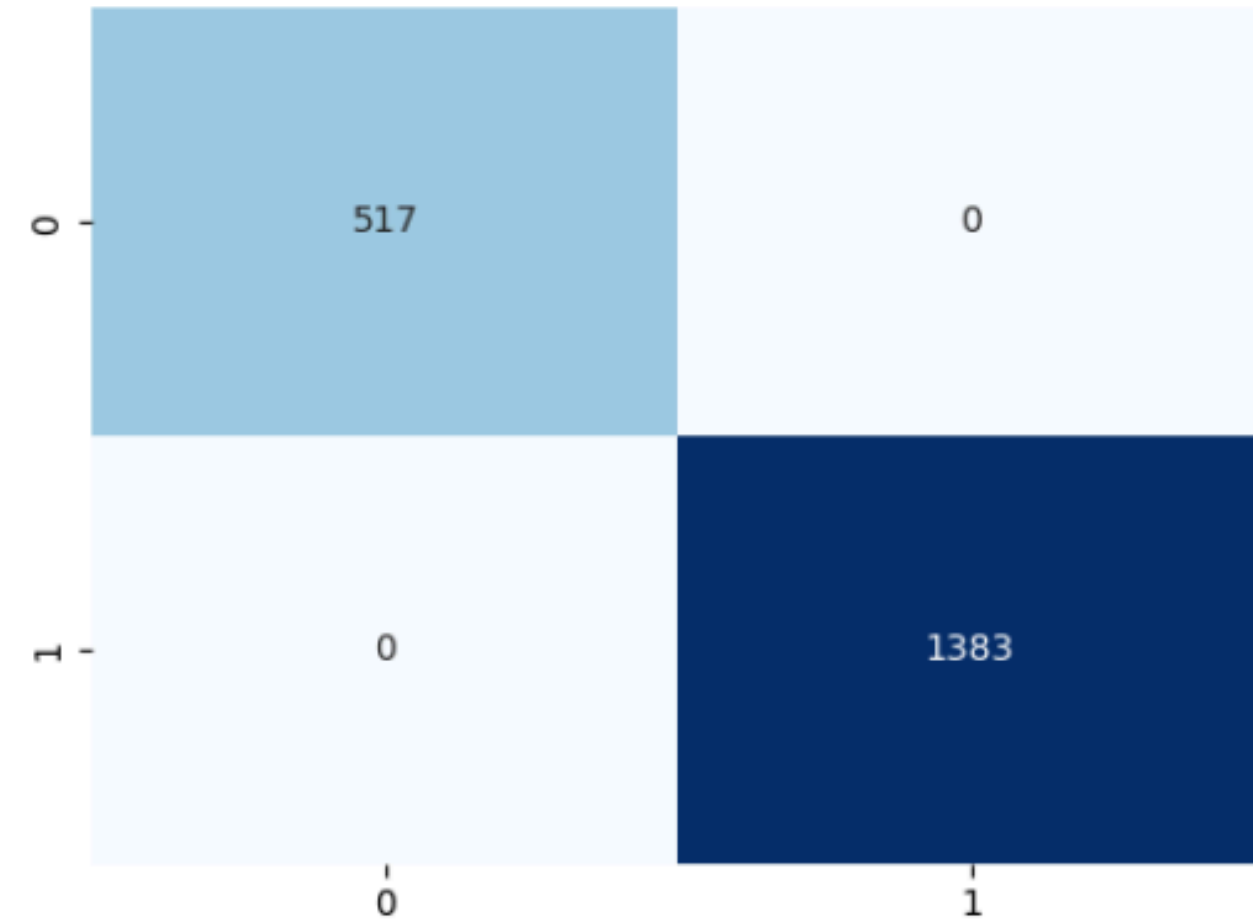
	review_score	has_review	target
1	4.0	1	1
14	5.0	1	1
22	5.0	1	1
24	1.0	1	0
29	1.0	1	0
36	5.0	1	1
45	5.0	1	1
49	1.0	1	0

Fixing Data Leakage

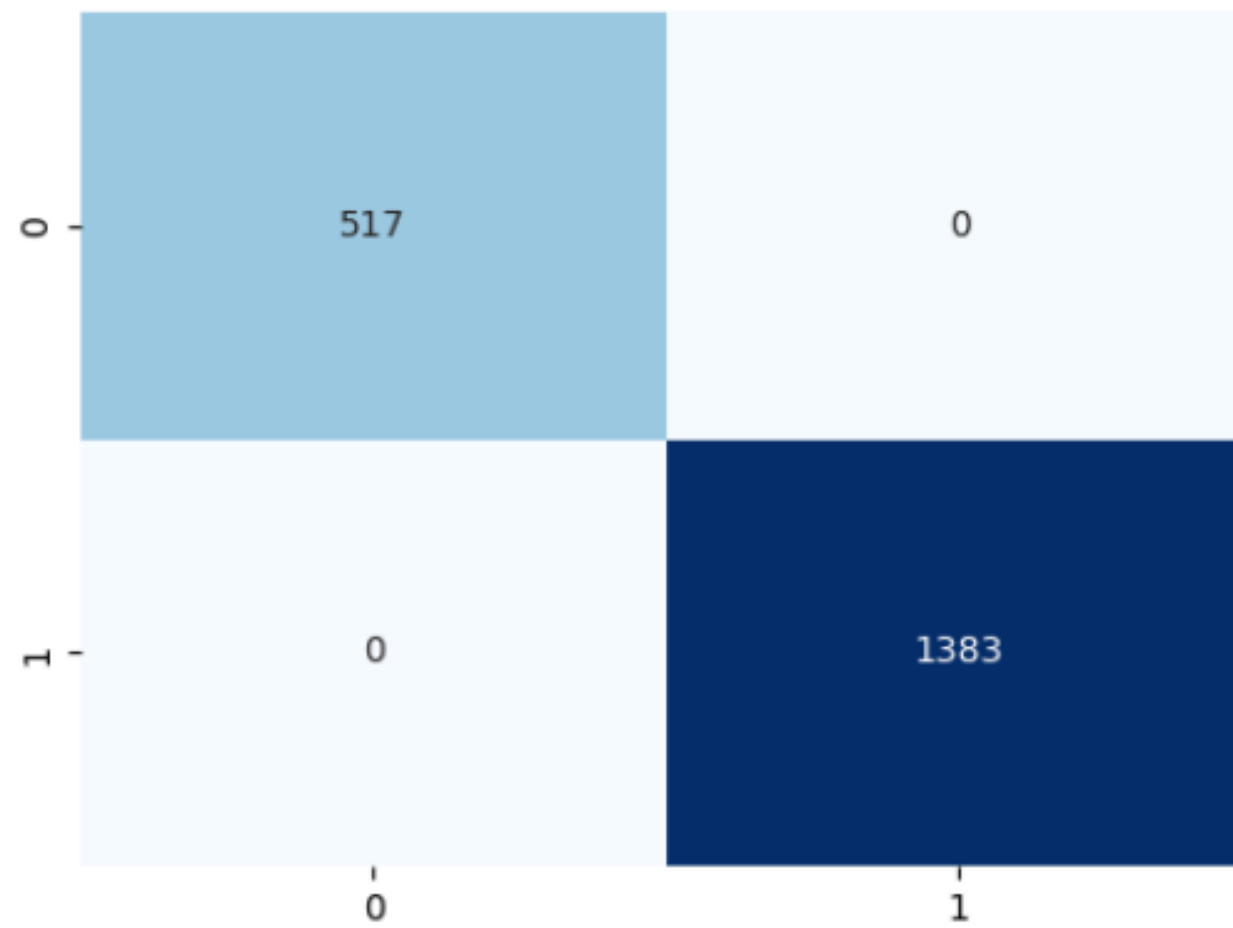
Logistic Regression (Leaked)



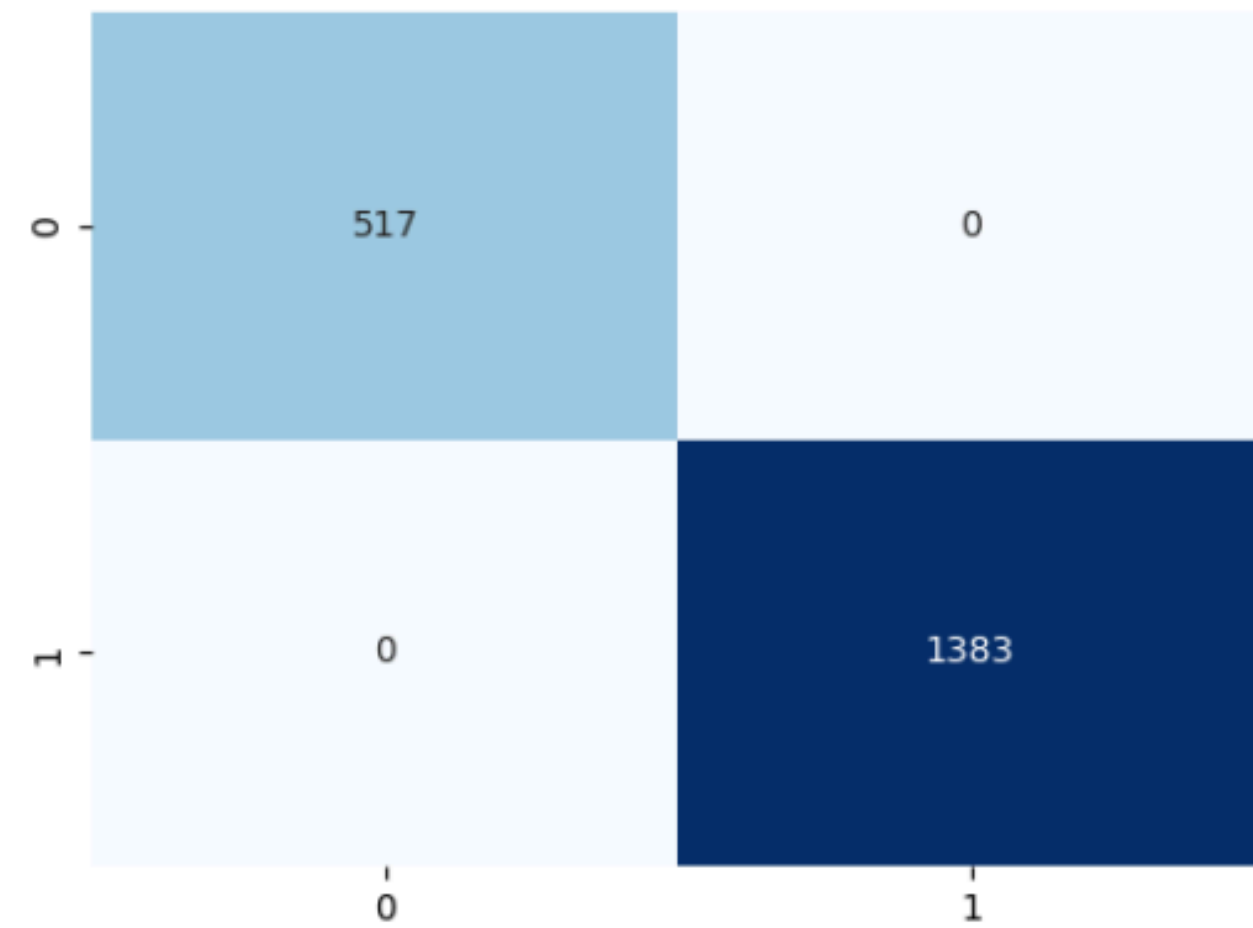
Decision Tree (Leaked)

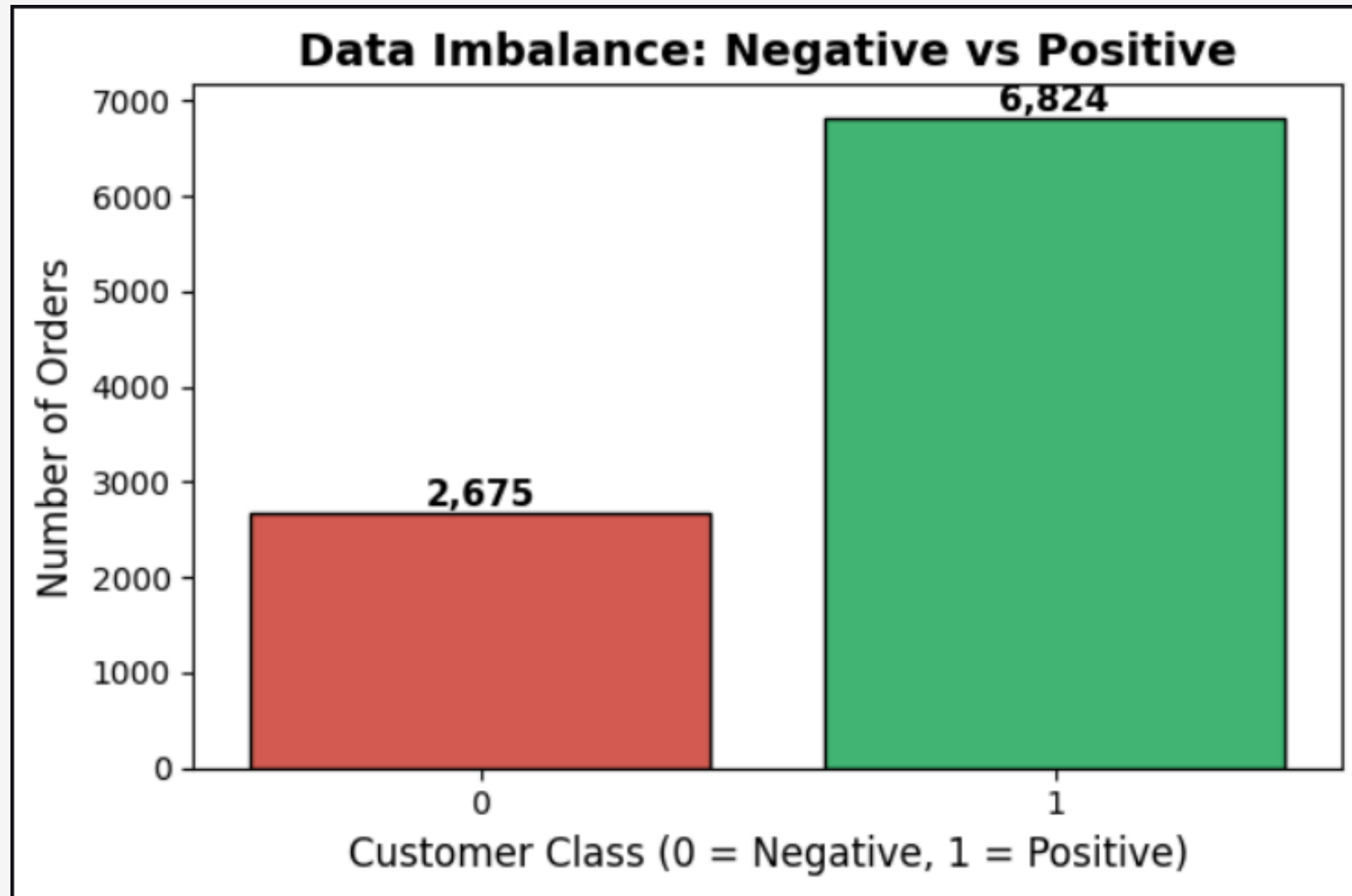


Random Forest (Leaked)



XGBoost (Leaked)



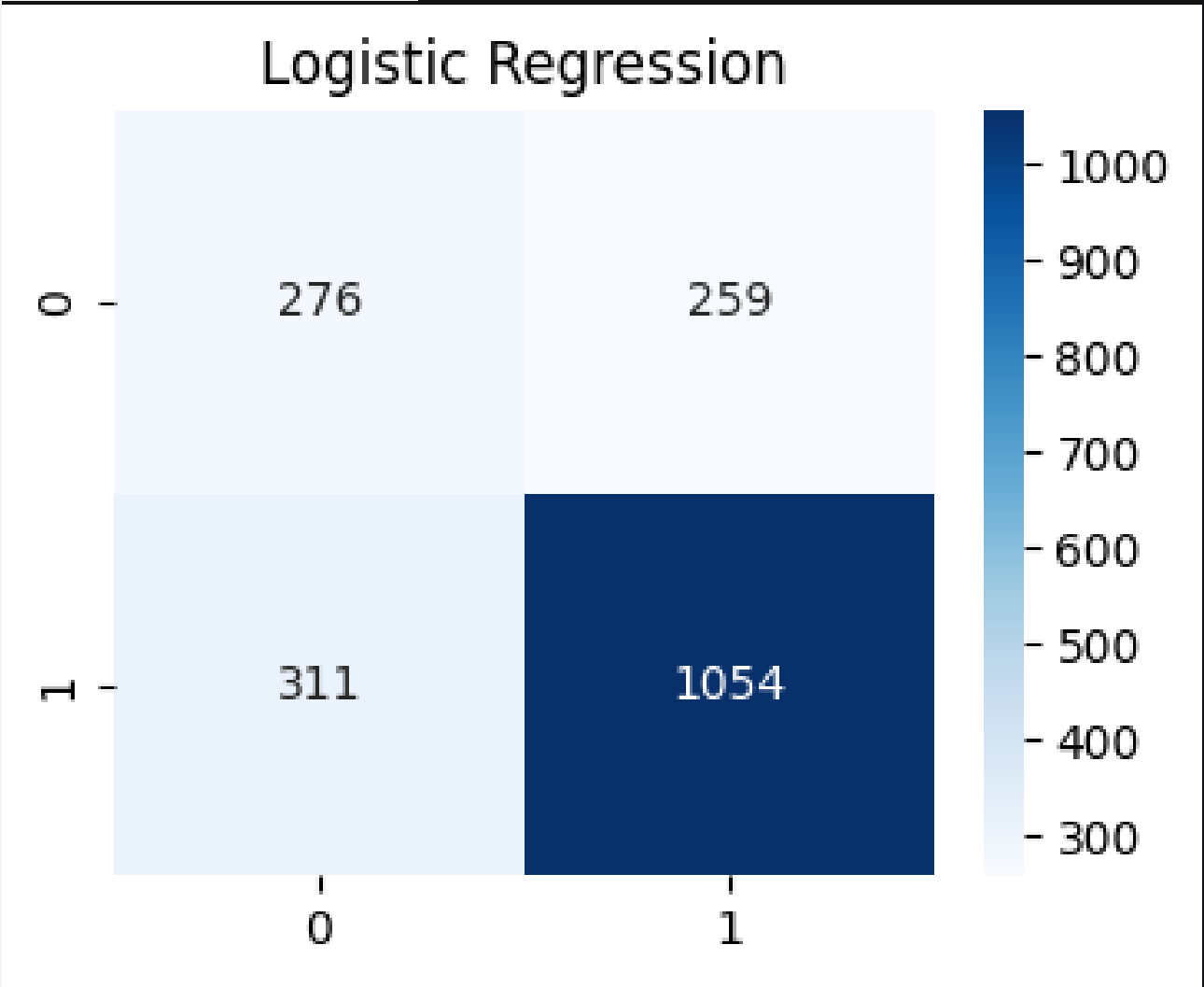
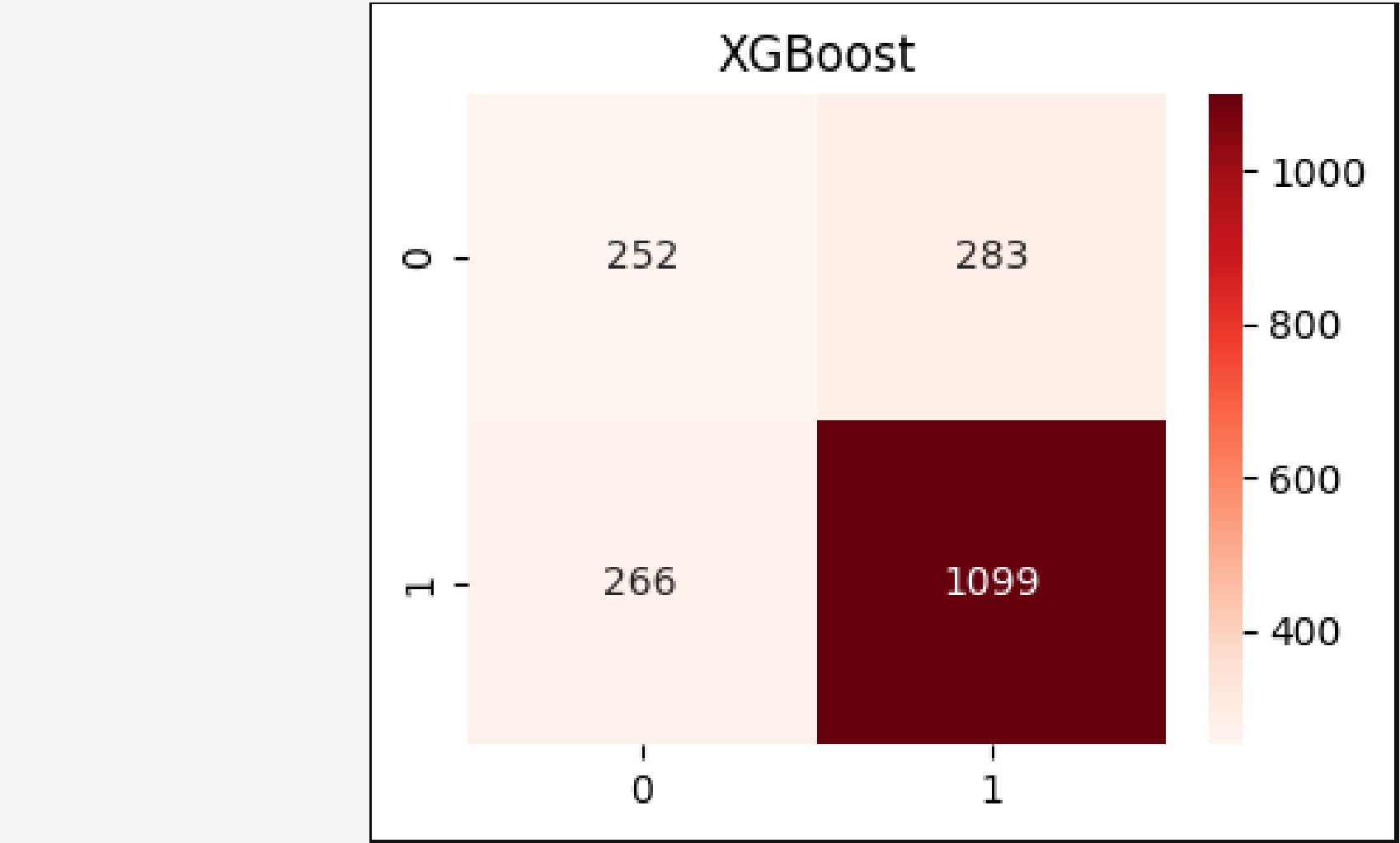
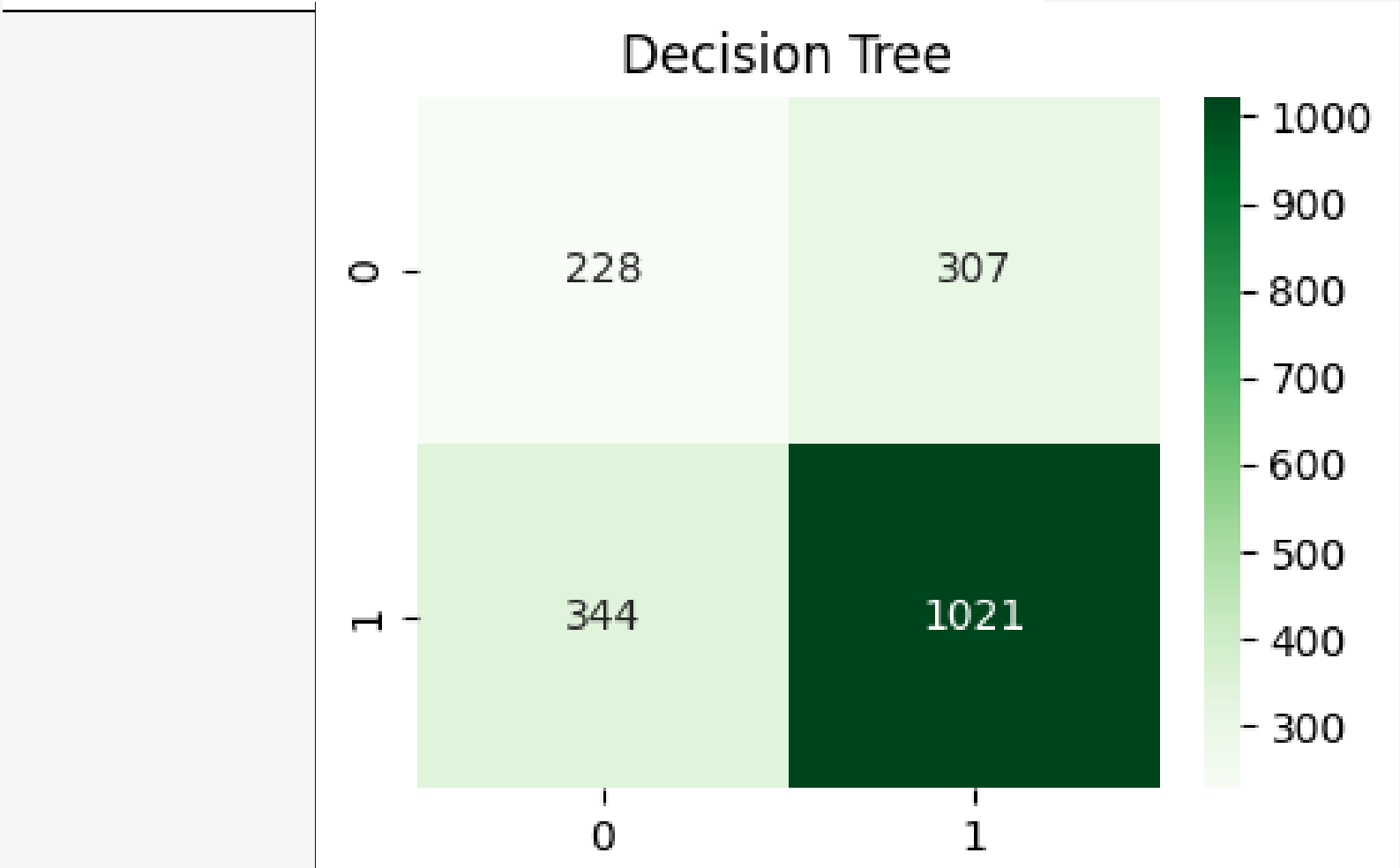
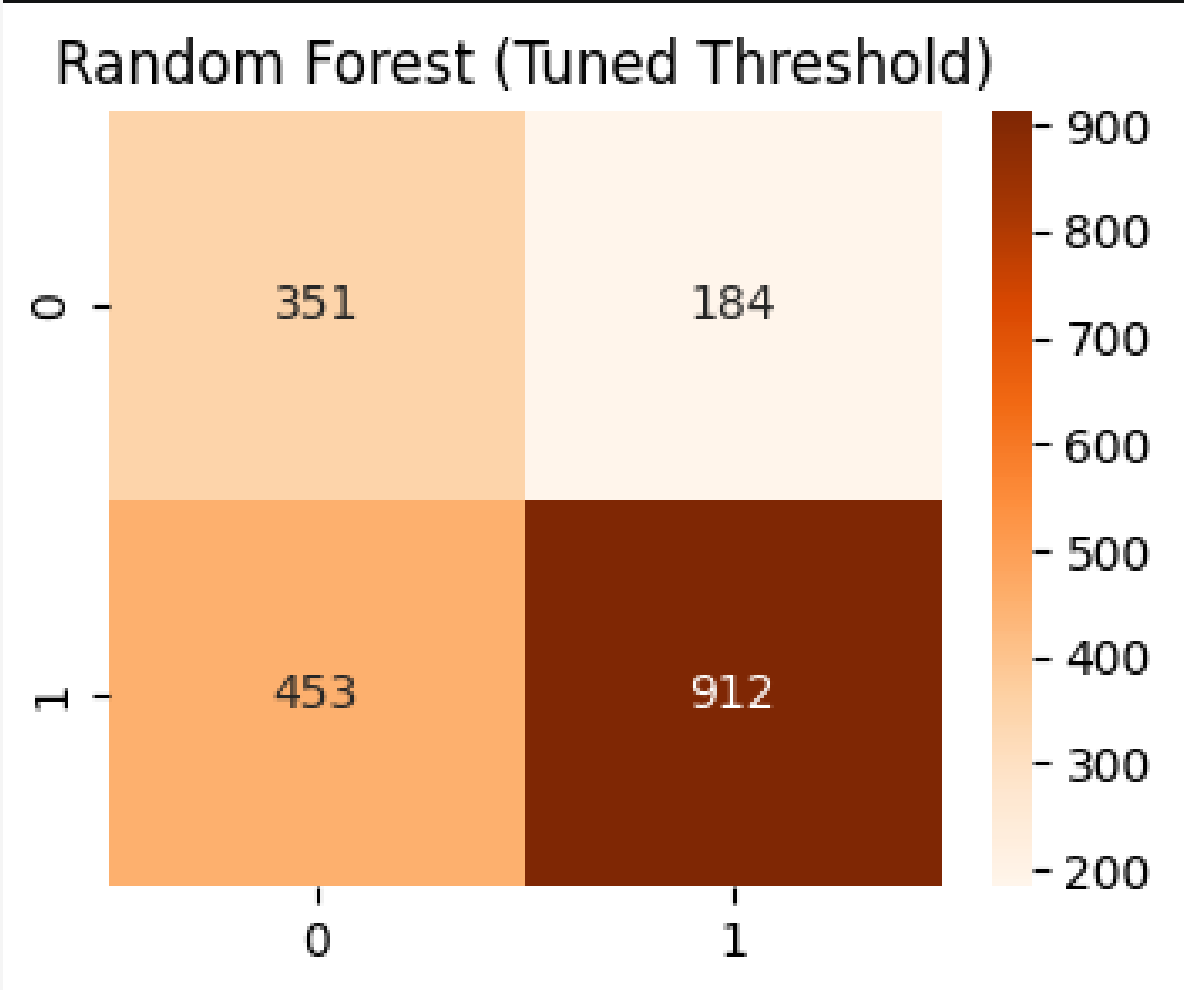


- **The Challenge:** Massive class imbalance. Models heavily favored the majority class (Positive)

- **Class Weighting:** Applied heavier penalties for misclassifying the minority class (Negative)

- **Threshold Tuning:** Lowered the decision threshold (to 0.75) to increase model sensitivity to potential risks

Model Performance
& Evaluation



Final Model Selection

--- FINAL MODEL PERFORMANCE COMPARISON ---

	Accuracy	Precision	Recall (Negative)	F1-Score	ROC-AUC
Random Forest	0.664737	0.832117	0.656075	0.741162	0.713322
Logistic Regression	0.700000	0.802742	0.515888	0.787155	0.687734
XGBoost	0.711053	0.795224	0.471028	0.800146	0.679180
Decision Tree	0.657368	0.768825	0.426168	0.758262	0.587077

- Selected Model: Random Forest (Tuned Threshold).

- Achieved the Highest Recall (Negative): 65.6%

- Strategic Trade-off: Sacrificed overall accuracy to maximize risk coverage, aligning perfectly with the Early Warning business objective

Part 4 : Visualization



Data

Review Score Distribution

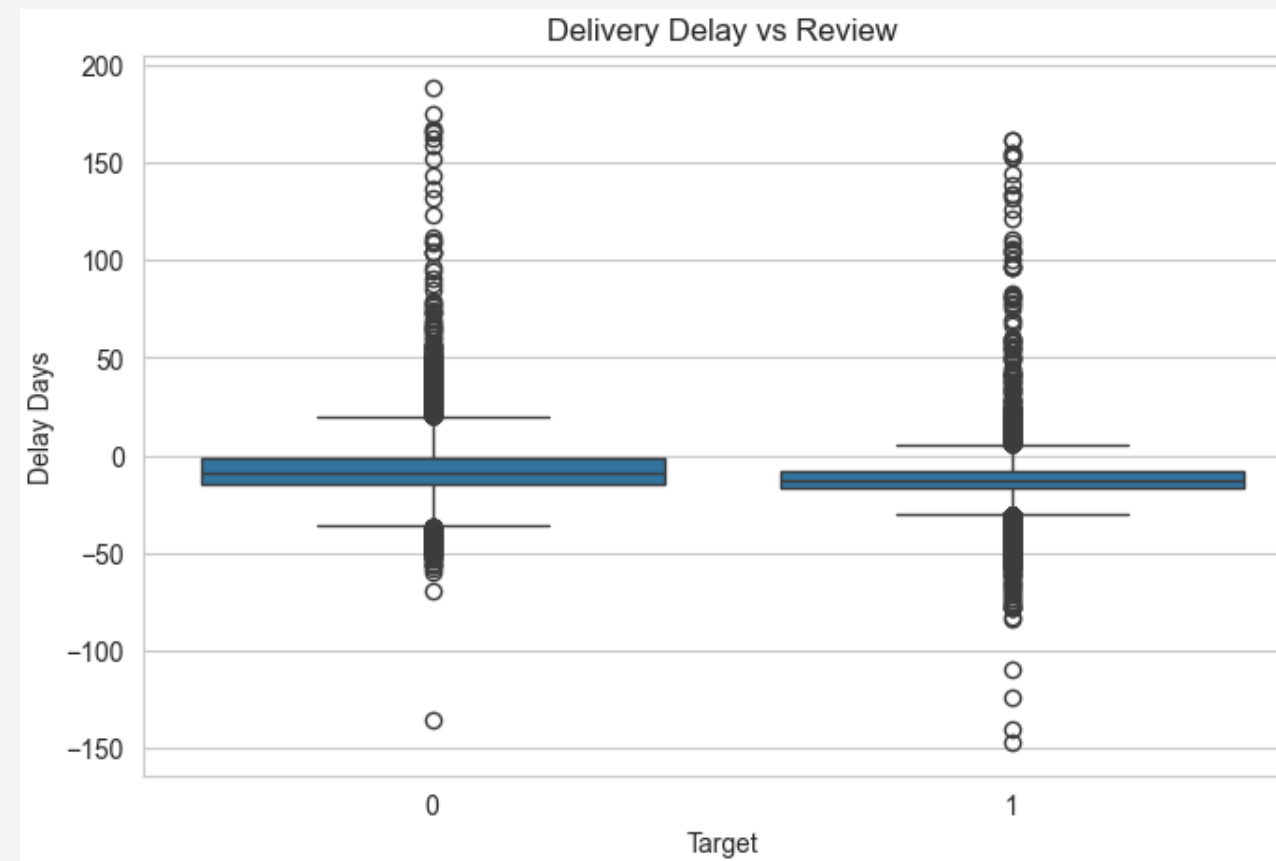
- Analyze the distribution of customer ratings.
- Identify the imbalance between positive and negative reviews.
-



Visualization

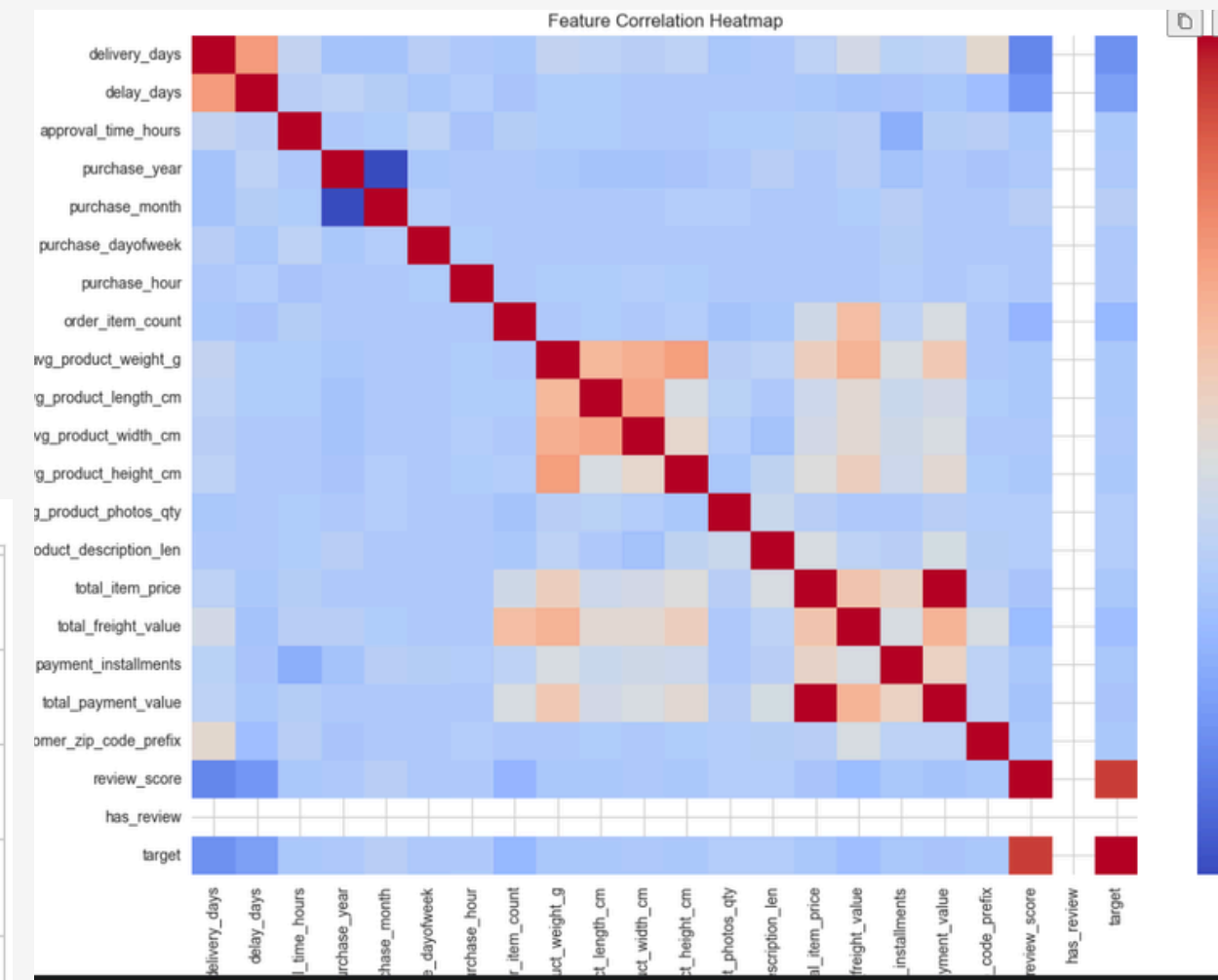
Delivery Delay Analysis

- Compare delivery delay between satisfied and dissatisfied customers.
- Delayed orders are more likely to receive negative reviews.



Correlation Analysis

- Explore relationships among numerical variables.
- Detect factors affecting customer satisfaction.



Model Visualization

Confusion Matrix

- Evaluate classification results.
- Measure correct and incorrect predictions.

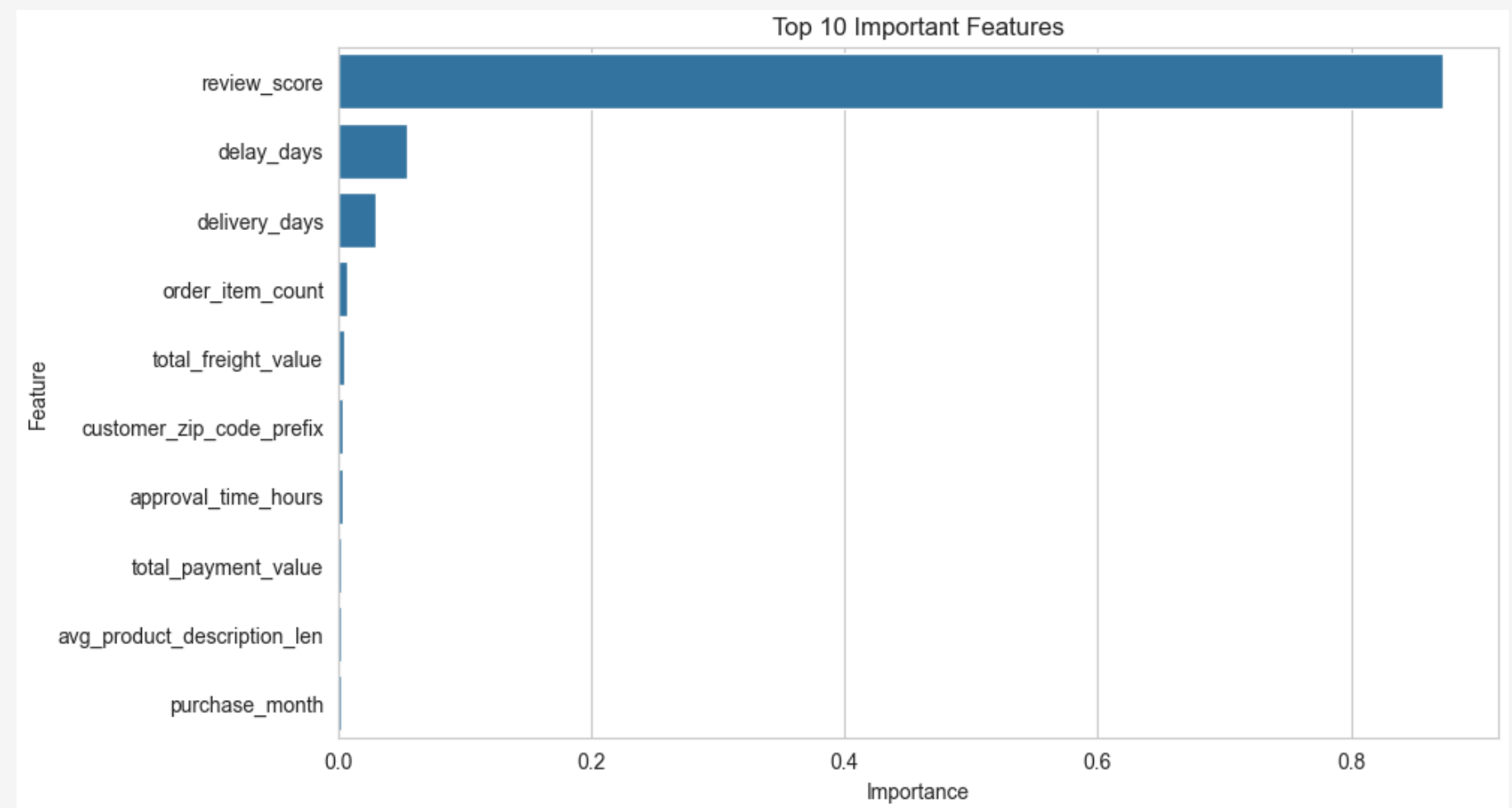
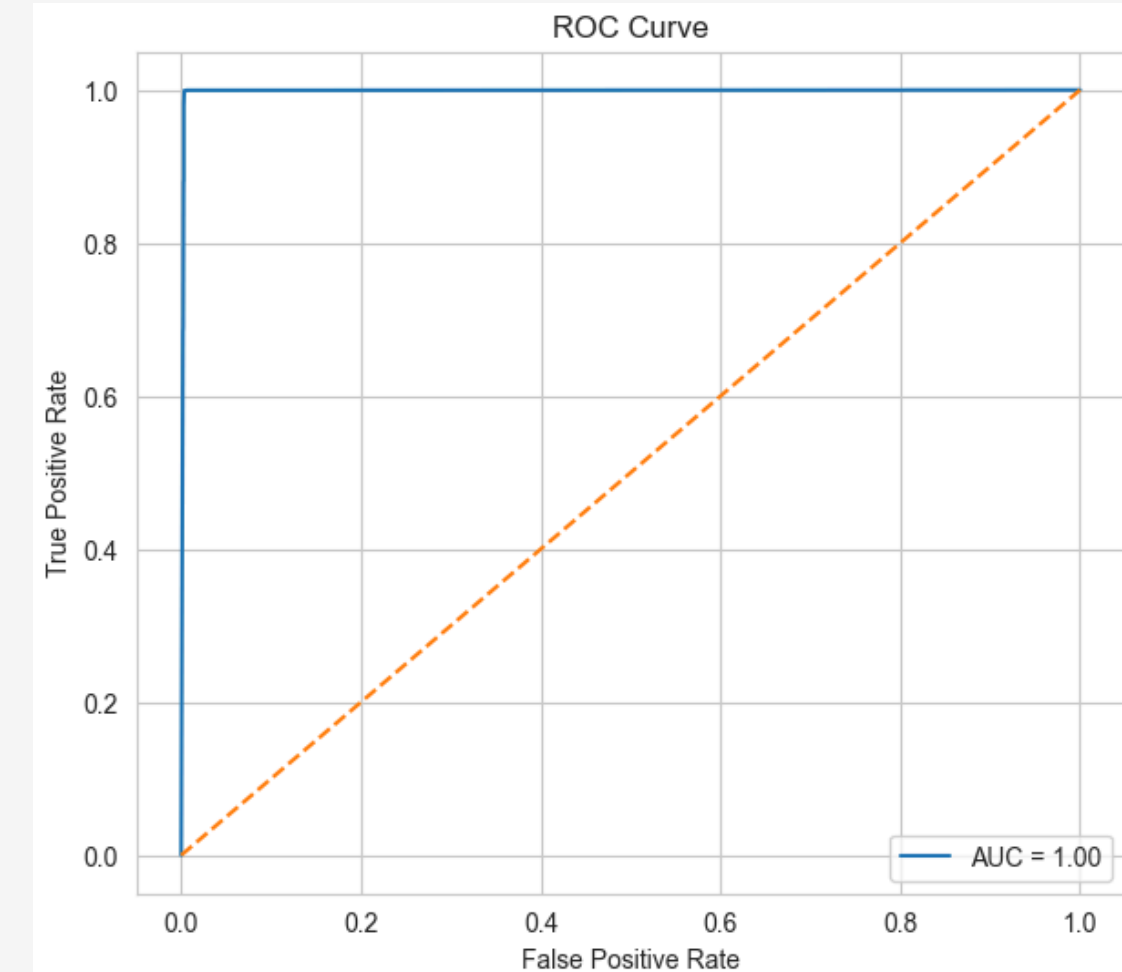
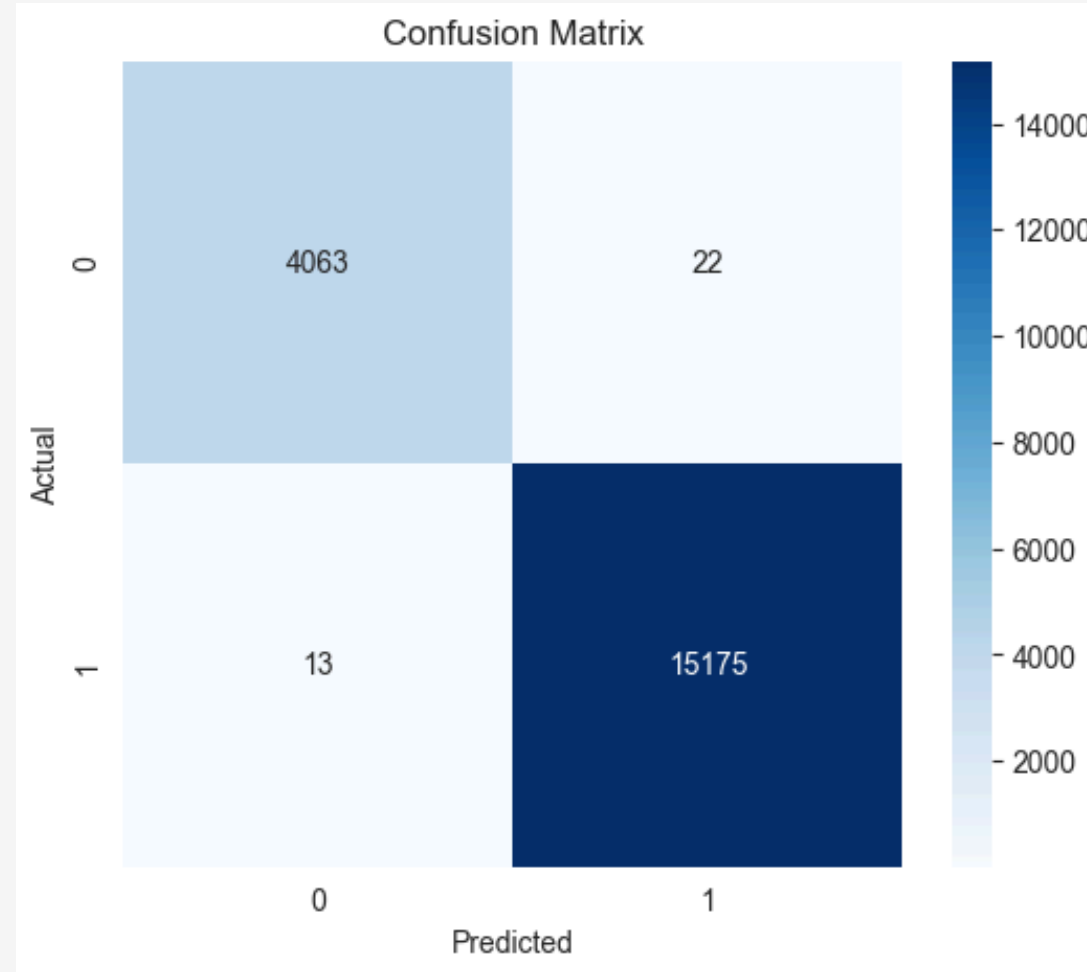
ROC Curve

- Assess the model's ability to distinguish between classes.
- Higher AUC indicates better performance.

Feature Importance

Identify the most influential features:

- Delay Days
- Delivery Time
- Shipping Ratio
- Product Volume
- Product Description Length



Key Achievements

- ✓ Successfully built an early warning system for customer review prediction.
- ✓ Performed data preprocessing and feature engineering.
- ✓ Compared multiple machine learning models.
- ✓ Evaluated model performance using visualization techniques.
- ✓ Identified the key factors influencing customer satisfaction.

Future Improvements

- ✓ Hyperparameter tuning
- ✓ SMOTE for handling imbalanced data
- ✓ LightGBM and CatBoost
- ✓ Real-time prediction system
-

Conclusion



THANK
YOU!

